JC4003 Natural Language Processing

Lecture 14: Automatic Speech Recognition and Text-to-Speech 2

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Outline

Automatic Speech Recognition (ASR)

Evaluate ASR models

Text-to-Speech (TTS) Approaches

Evaluate TTS models

Automatic Speech Recognition (evaluation)

Word Error Rate (WER)

Evaluating ASR models involves comparing the model's transcriptions with reference (ground truth) transcripts. Key metrics include Word Error Rate (WER), which measures the percentage of errors (**S**ubstitutions, **D**eletions, and **I**nsertions) in the transcribed text.

D

Ground true: How are you today John

ASR output: How you a today Jones

I S

Word Error Rate (WER, cont.)

The word error rate (WER) is then defined as follows. The lower the WER score, the better the ASR model performs.

NB: because the equation includes insertions, WER can be greater than 100%.

$$\mathrm{WER} = \frac{S + D + I}{N} \times 100\%$$

- S stands for substitutions
- I stands for insertions
- D stands for deletions
- N is the number of words in the reference

Word Error Rate (WER, cont.)

Consider the example from CallHome corpus:

ASR Output: i got it to the fullest i love to protable form of stores last night

Ground true: i um the phone is i left the protable phone upstairs last night

To calculate WER, we first need to align the ASR output with the ground truth (i.e., the reference).

REF	i	*	*	um	the	phone	is	i	left	the	portable	*	phone	upstairs	last	night
ASR	i	got	it	to	the	*	fullest	i	love	to	portable	form	of	stores	last	night
error		I	I	S		D	S		S	S		I	S	S		

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Word Error Rate (WER, cont.)

This utterance has 6 substitutions (S=6), 3 insertions (I=3), and 1 deletion (D=1).
The number of words in the reference is 13 (N=13).

Thus, in this example,

$$\text{WER} = \frac{S + D + I}{N} \times 100\% = \frac{6 + 1 + 3}{13} \times 100\% = 76.9\%$$

REF	i	*	*	um	the	phone	is	i	left	the	portable	*	phone	upstairs	last	night
ASR	i	got	it	to	the	*	fullest	i	love	to	portable	form	of	stores	last	night
error		I	I	S		D	S		S	S		I	S	S		

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Alignment In WER

Alignment is crucial for calculating Word Error Rate (WER) as it ensures accurate identification of error types (substitutions, deletions, insertions) between the model output and reference text.

The optimal alignment minimises the error count. Since the word count in the model output and the reference text may not always be the same, a simple comparison could result in incorrect calculations.

Optimal alignment ensures that only the necessary substitutions, deletions, and insertions are considered, leading to a more accurate WER calculation.

Levenshtein Distance

Levenshtein distance, also known as edit distance, measures the minimum number of substitutions, insertions, or deletions needed to convert one string into another.

It is commonly used in natural language processing and ASR evaluation to quantify the difference between two sequences, e.g. in WER and spell check.

WER:

		D			
Ground true:	How	are	you	today	John
ASR output:	How	you	a	today	Jones
			I		S

Spell check:

			D				
String 1:	_	F	L	O	A	T	
String 2:	_	B		O	A	T	S
		S					I

Text-to-Speech (TTS)

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First Speech Synthesis Machines

Text-to-Speech (TTS) has been around even longer than speech recognition. Between 1769 and 1790, Wolfgang von Kempelen built the first machine that could create "full sentences".

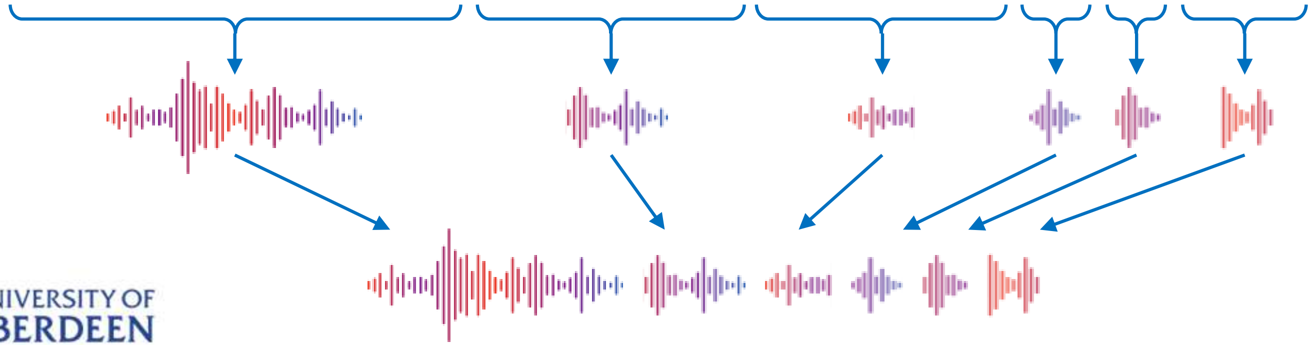
It worked by using bellows for lungs, a rubber mouth, and different parts to copy speech sounds. The operator could move these parts by hand to produce different consonants and vowels.

Concatenative TTS System

Concatenative TTS systems convert text into speech by stitching together pre-recorded word voice. The system splits texts into words, retrieves the corresponding audio, and joins them to produce full sentences.

While this method offers high-quality speech, it can result in less fluid outputs and struggles with words not present in their database.

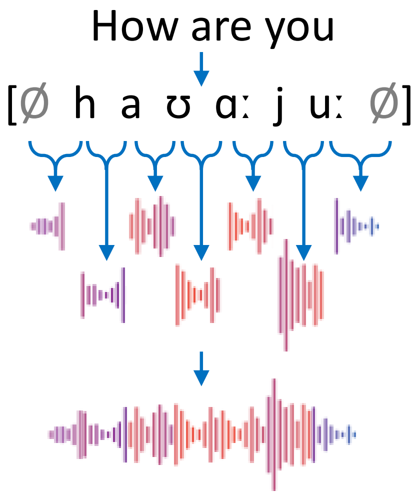
Your account balance is one thousand two hundred and ten dollars.



Diphone Concatenation TTS System

Diphone Concatenation TTS systems generate speech by combining pre-recorded diphones, offering more natural transitions than rule-based methods. While using real recordings gives a human-like sound, transitions can still feel less smooth, especially in complex speech.

A diphone refers to a segment of speech that spans from the middle of one phoneme to the middle of the next.



My cat who lives dangerously had nine lives.

He wanted to go for a drive in the country

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Formant Synthesis-based TTS System

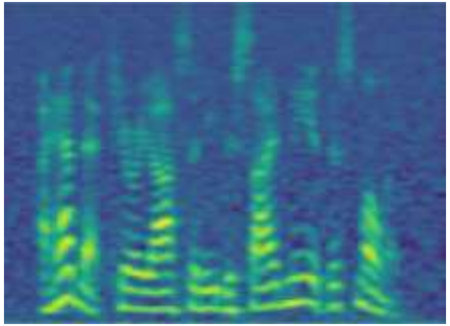
Formant synthesis-based TTS systems use mathematical models to mimic human speech sounds, like pitch and frequency, allowing for precise control over voice characteristics. While flexible and easy to implement, the resulting speech often sounds mechanical and lacks the natural quality found in modern deep learning-based methods.

How are you

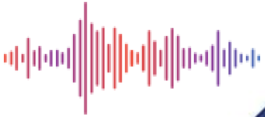
[h a ʊ ɑː j uː]

$f_{h,2}$	$f_{a,2}$	$f_{ʊ,2}$	$f_{ɑː,2}$	$f_{j,2}$	$f_{uː,2}$
$f_{h,1}$	$f_{a,1}$	$f_{ʊ,1}$	$f_{ɑː,1}$	$f_{j,1}$	$f_{uː,1}$
$f_{h,0}$	$f_{a,0}$	$f_{ʊ,0}$	$f_{ɑː,0}$	$f_{j,0}$	$f_{uː,0}$

vocoder



FFT



E.g. Good afternoon radio audiences

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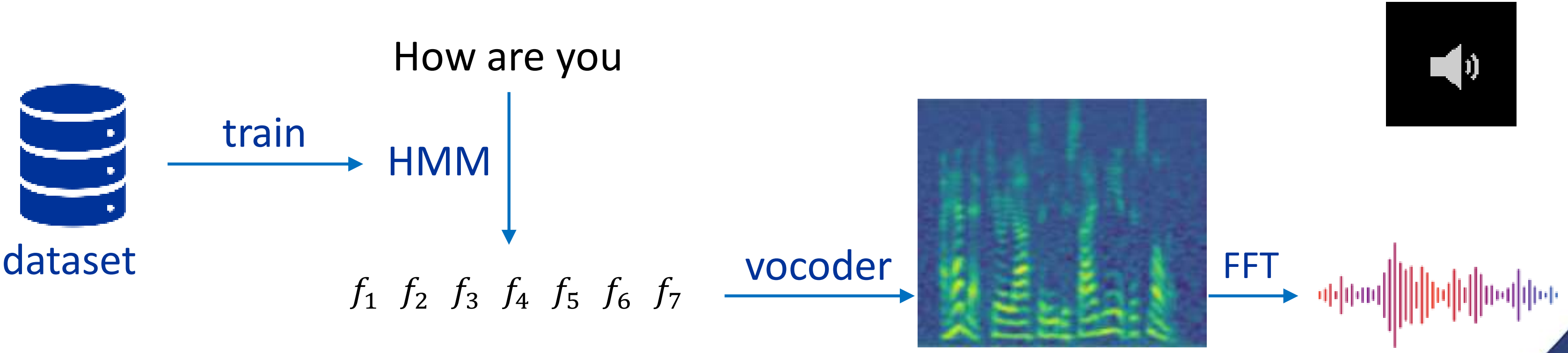
Statistical Parametric Synthesis-based TTS

Statistical parametric TTS generates speech by learning from large datasets, typically using Hidden Markov Models (HMMs) to predict speech parameters like pitch and duration. While flexible and able to handle different styles, the resulting speech often sounds mechanical and lacks naturalness.



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Neural Network-based TTS

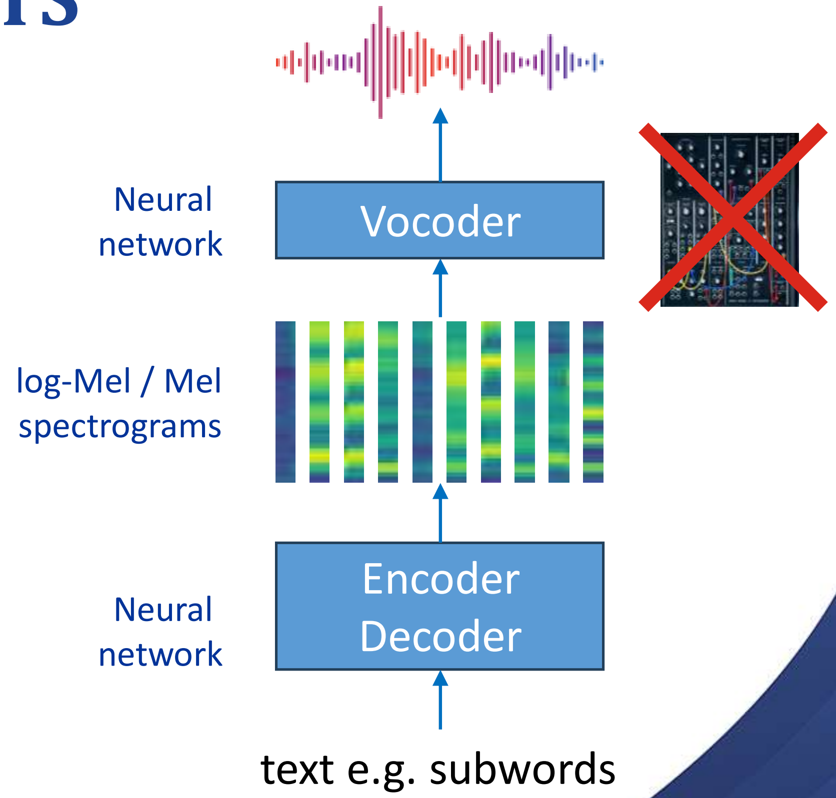
E.g. Edge TTS

Neural Network-based TTS

Neural network-based TTS (e.g. Edge TTS) generates natural and fluent speech using deep learning models. These models such

1. create a spectrogram from the text
2. use a neural vocoder to produce the speech.

Compared to traditional methods, neural network-based TTS offers significantly better sound quality and naturalness, capturing richer speech features and supporting multiple languages and emotions.

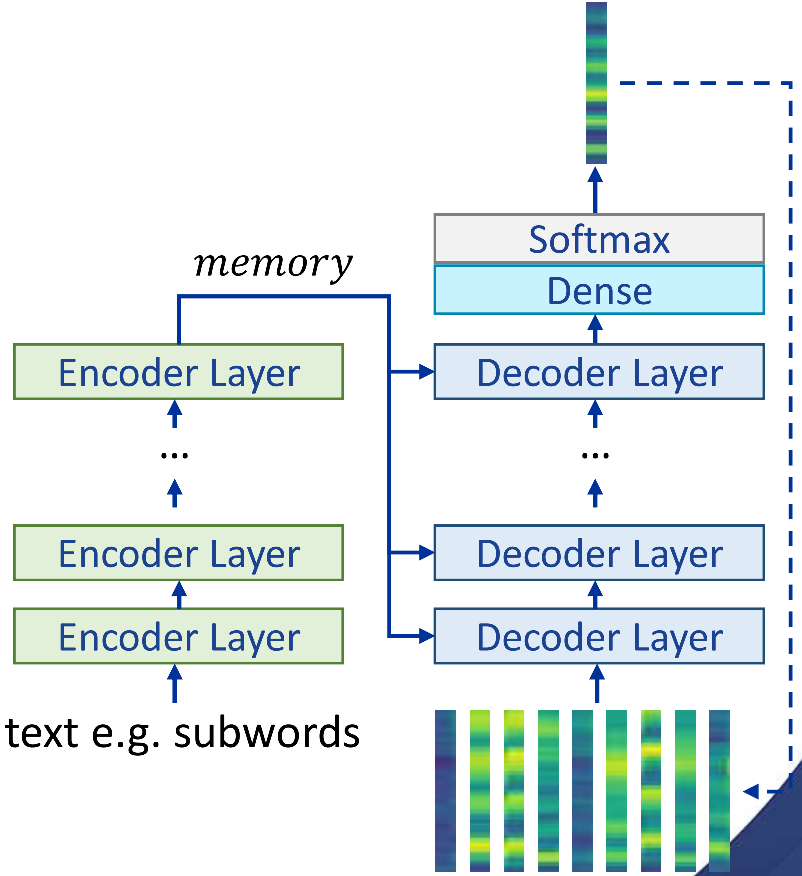


Text-to-Mel

Edge-TTS uses a Transformer-based encoder-decoder model to convert text into Mel-spectrograms.

The encoder processes the text into high-dimensional representations, capturing its semantics and intonation.

The decoder then predicts the Mel-spectrograms frame by frame, utilising autoregressive or teacher forcing to produce Mel-spectrograms.



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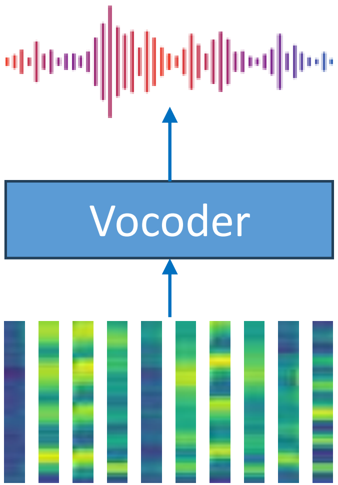
Vocoder

A vocoder is used to synthesise speech by analysing and modifying audio signals, converting intermediate representations like Mel-spectrograms into audible waveforms.

Neural vocoders (e.g. **WaveNet**) improved the quality and naturalness of synthesised speech by generating high-resolution audio directly from these features.

Neural vocoders play a critical role in ensuring that speech output is smooth and natural.

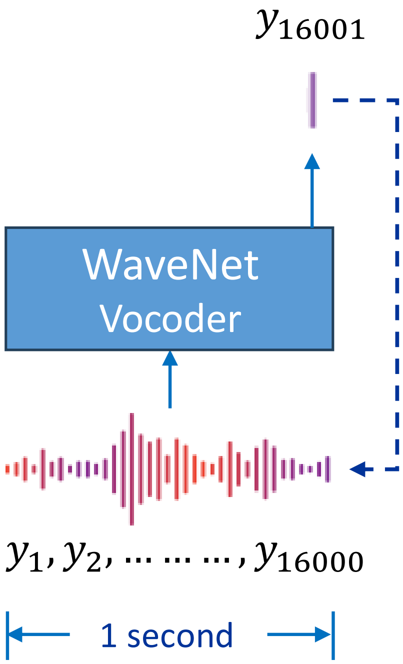


log-Mel / Mel spectrograms etc.

Vocoder -- WaveNet

WaveNet is a deep generative vocoder. It operates through an autoregressive process, like transformers, predicting each audio sample point based on previous samples, enabling it to generate speech, music, or other audio signals.

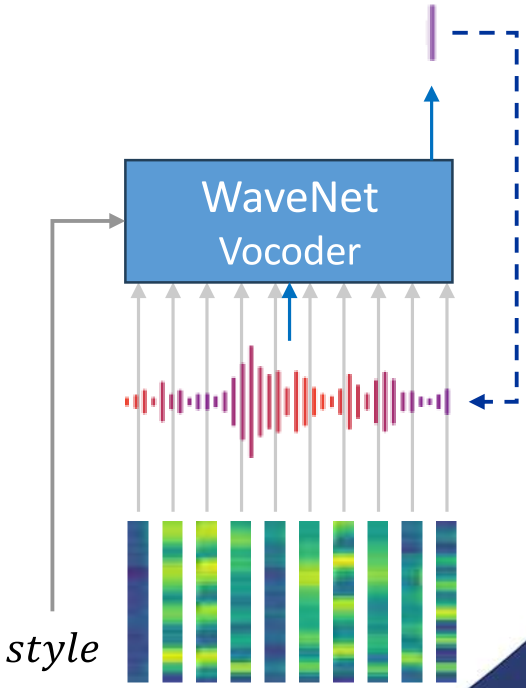
WaveNet directly models the raw waveform, enabling it to produce highly expressive and natural audio. However, this approach also requires WaveNet to generate a large number of output vectors at each time step, particularly given the high sampling rates required for natural-sounding audio.



Vocoder -- WaveNet

WaveNet also supports generating the final waveform based on given mel spectrograms. In this case, the mel spectrograms can be used as additional inputs (conditioning vectors) to guide the model, conditioning the network to produce audio that matches the spectrogram's characteristics.

WaveNet can also accept globally conditioning vector representations to guide the stylistic characteristics of the generated audio. These global conditions allow the model to adjust features such as voice style, tone, or even the identity of the speaker, providing greater control over the output.



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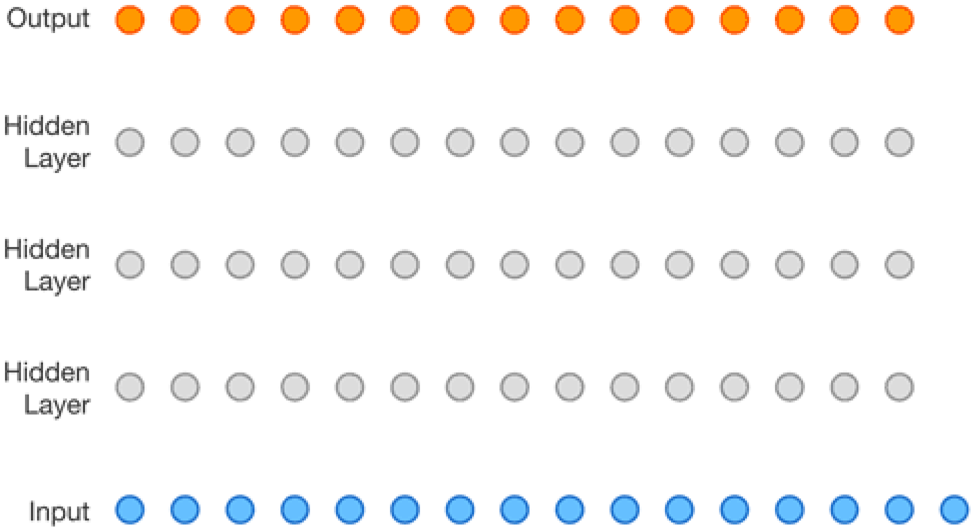
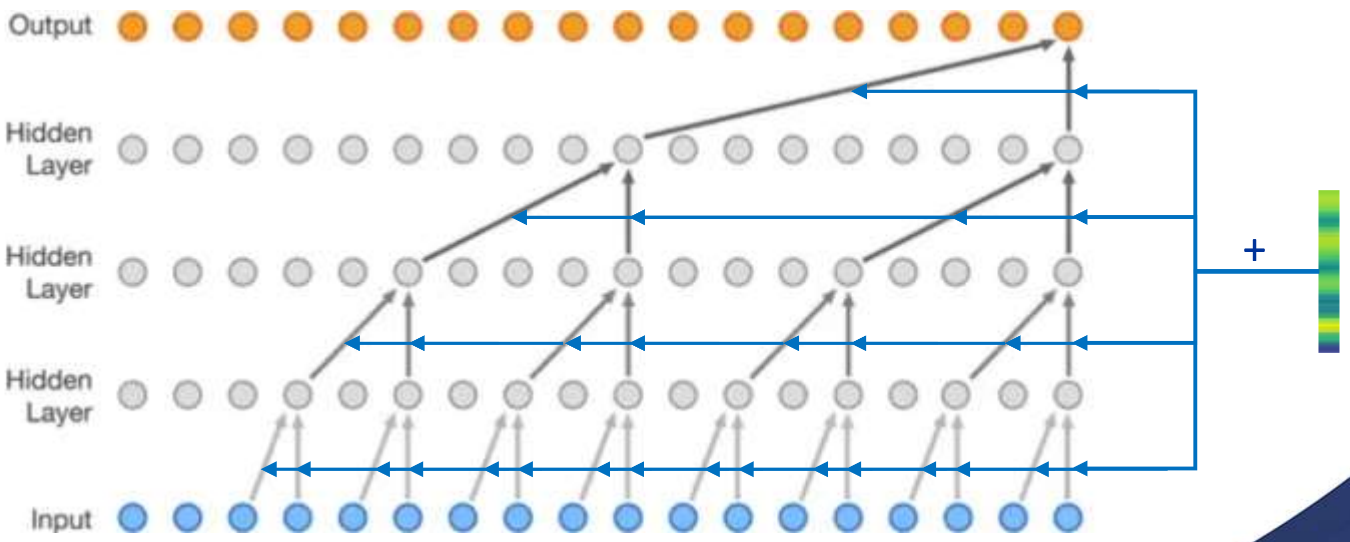
Vocoder – WaveNet (cont.)

WaveNet utilises causal convolutions internally to ensure that the generated audio adheres to the correct temporal sequence, which is crucial for autoregressive models. Because each time step's output can only depend on the current and previous inputs, it supports parallel training by adopting teacher forcing (like transformers).



Vocoder – WaveNet (cont.)

In WaveNet, both global and local conditioning vectors are incorporated into each layer of the causal convolutions, by 1x1 conv operations. Through adjustments in the convolutional layer weights, the conditioning vectors influence the audio generation process.



Neural Network-based TTS Performance

Neural network-based TTS systems have greatly improved speech synthesis by generating more natural and expressive speech. Models like Edge TTS WaveNet capture complex speech patterns, producing intelligible, fluid, and natural-sounding voices. These systems outperform traditional methods in adaptability to different voices, emotions, and languages.

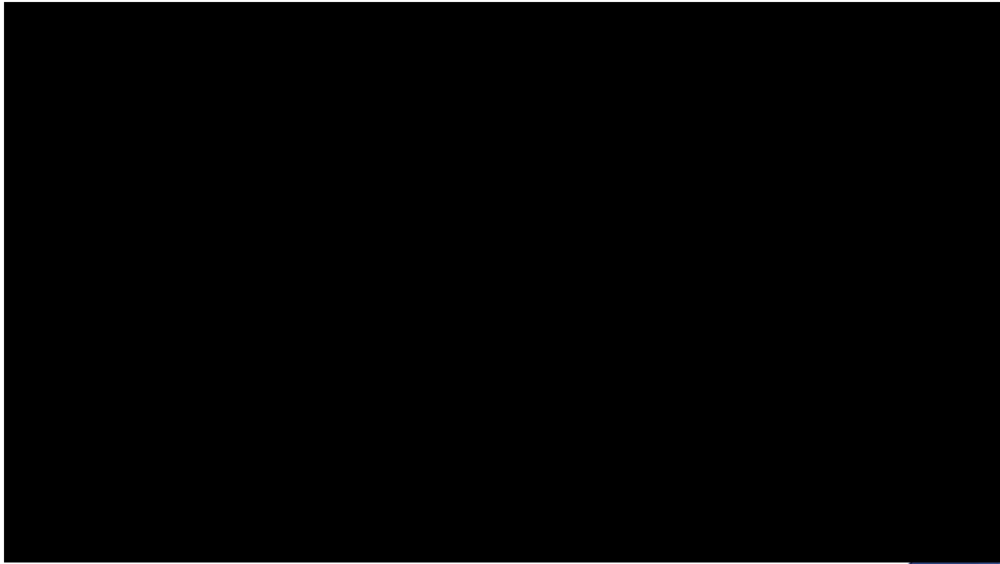
Chinese TTS example by Edge TTS

Chinese TTS example by Edge TTS with background music

Applications in Music and Singing Synthesis

In addition to generating speech, Text-to-Speech technology plays a significant role in other areas such as music and singing synthesis. Examples include singing voice synthesis and music composition.

Much of this work relies on voice banks, which involve collecting a large amount of pre-recorded sounds that are then reassembled and edited to generate audio. For instance, Vocaloid operates on this principle.



Hatsune Miku is synthesised using a Voicebank, which is based on a collection of pre-recorded voice samples.

Applications in Music and Singing Synthesis

Popular deep learning or large model-based approaches have yet to match the performance of Voicebank-based methods in these areas, but it seems that this gap may not be far from closing.

TTS Evaluation

TTS Evaluation

Evaluating the performance of Text-to-Speech (TTS) models involves a variety of methods and metrics, broadly divided into subjective and objective assessments. Below are the commonly used evaluation methods:

- Subjective evaluation involves human listeners rating or comparing the synthesised speech, focusing on its naturalness, clarity, and overall user experience. While time-consuming, it provides a direct reflection of the model's real-world effectiveness.
- Objective evaluation uses automated metrics that do not require human input. While it doesn't fully capture the naturalness of speech, it is useful for model optimisation and rapid evaluation.

MOS (Mean Opinion Score)

MOS is one of the most widely used subjective metrics in TTS evaluation. Listeners rate speech samples on a scale of 1 to 5:

- 5: Very natural
- 4: Close to natural
- 3: Fairly acceptable
- 2: Noticeably synthetic
- 1: Completely unacceptable

Listeners score the speech based on its naturalness, fluency, and how human-like it sounds, with the final score being the average across all listeners.

MOS (Mean Opinion Score, cont.)

When using MOS to evaluate TTS systems, several key considerations are essential to ensure the reliability and accuracy of the evaluation results:

Participant Background: Ensure that participants have adequate auditory capabilities, and they be fluent in the target language to properly assess the speech.

Sample Diversity: Use a diverse group of participants to reduce subjective bias. Participants should come from a variety of age groups, genders, and cultural backgrounds.

Consistency in Scoring Criteria: Ensure that participants clearly understand the scoring criteria, such as naturalness, clarity, and pitch, to avoid different interpretations of the same score.

Clear Scoring Range: Clearly define what each score means (e.g., a score of 5 means “very natural,” while 1 means “completely unacceptable”) to avoid subjective bias in scoring.

A/B Testing

A/B testing is a common subjective evaluation method for assessing TTS models. It involves presenting listeners with two different speech samples (A and B) and asking them to choose which one sounds more natural or closer to human speech.

Steps of A/B Testing:

1. Select two speech samples (A and B), potentially from different TTS models.
2. Listeners hear both samples.
3. They choose the better sample based on criteria such as naturalness and fluency, or they can opt for "no preference."
4. Feedback is collected and the preference percentages are calculated.

The listeners are often unaware of the source of the samples (e.g., different models or versions of the same model), helping to reduce bias.

PESQ (Perceptual Evaluation of Speech Quality)

PESQ is a standardised method for objectively assessing speech quality, originally developed to evaluate the quality of speech codecs and communication systems. It measures the difference in quality between synthesised or transmitted speech and a reference speech signal by simulating human auditory perception, resulting in a score that reflects the perceived quality of the speech.

PESQ compares a reference speech signal with a processed speech signal, using auditory models to calculate the differences in speech quality. It breaks the audio into multiple frequency bands and assesses the distortion in each band.

Finally, PESQ provides a score ranging from 1 to 4.5, with higher scores indicating better perceived speech quality.

PESQ (Perceptual Evaluation of Speech Quality)

Advantages

- PESQ offers a fast, repeatable, and automated evaluation method, making it widely used for speech quality assessment, particularly in communication and synthesis systems.

Disadvantages

- As an objective evaluation, PESQ may not fully capture human subjective experience, especially for tasks involving complex emotional or natural speech synthesis.

STOI (Short-Time Objective Intelligibility)

STOI is an objective measure used to assess speech intelligibility, particularly in noisy or processed conditions. It compares the spectral differences between reference and processed speech over short segments (typically 384 ms).

- Short-time analysis evaluates intelligibility by comparing speech in brief time windows.
- Effective in noisy environments, commonly used for assessing speech in telecommunication or noise suppression applications.
- Scores range from 0 to 1, where 1 indicates perfect intelligibility.

Conclusion

ASR Evaluation: Word Error Rate (WER) is the key metric for evaluating ASR models, quantifying transcription errors in terms of substitutions, deletions, and insertions.

TTS Methods: Covered various TTS approaches, from concatenative and formant synthesis to advanced neural network-based methods like Edge TTS and WaveNet.

TTS Evaluation: Discussed subjective metrics like MOS and A/B testing, and objective metrics like PESQ and STOI to assess speech quality and intelligibility.

Advancements in TTS: Neural network-based TTS systems significantly improve the naturalness and fluidity of synthesised speech, outperforming traditional methods in versatility and quality.